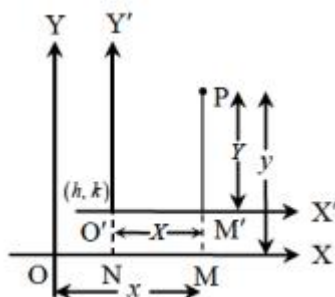


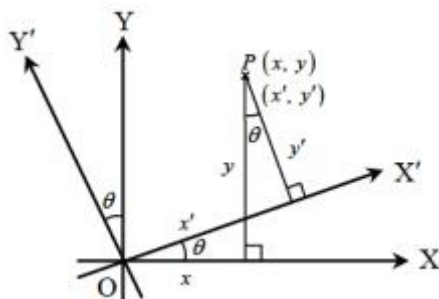
STRAIGHT LINE

- **Distance Formula:** The distance between two points (x_1, y_1) and (x_2, y_2) is given by $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.
- **Section Formula:**
 - (1) **Internal Division:** If a point (x, y) divides the line joining (x_1, y_1) and (x_2, y_2) internally in the ratio $m : n$, then $x = \frac{mx_2 + nx_1}{m + n}$, $y = \frac{my_2 + ny_1}{m + n}$.
 - (2) **External Division:** If a point (x, y) divides the line joining (x_1, y_1) and (x_2, y_2) externally in the ratio $m : n$, then $x = \frac{mx_2 - nx_1}{m - n}$, $y = \frac{my_2 - ny_1}{m - n}$.
 - (3) **Mid point:** If a point (x, y) is the midpoint of the line joining (x_1, y_1) and (x_2, y_2) , then $x = \frac{x_2 + x_1}{2}$, $y = \frac{y_2 + y_1}{2}$.
- **Area of triangle:** Area of a triangle with three vertices (x_1, y_1) , (x_2, y_2) and (x_3, y_3) is given by $Area = \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = \frac{1}{2} [x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)]$.
- **Collinearity of three points:** Three points (x_1, y_1) , (x_2, y_2) and (x_3, y_3) are collinear if area bounded by them is zero. i.e. $\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0$.
- **Area of a Quadrilateral:** Area of quadrilateral with four vertices (x_1, y_1) , (x_2, y_2) , (x_3, y_3) and (x_4, y_4) is $Area = \frac{1}{2} \left[\begin{vmatrix} x_1 & y_1 \\ x_2 & y_2 \end{vmatrix} + \begin{vmatrix} x_2 & y_2 \\ x_3 & y_3 \end{vmatrix} + \begin{vmatrix} x_3 & y_3 \\ x_4 & y_4 \end{vmatrix} + \begin{vmatrix} x_4 & y_4 \\ x_1 & y_1 \end{vmatrix} \right]$.
- **Area of a Polygon:** Area of polygon with n vertices (x_1, y_1) , (x_2, y_2) , (x_3, y_3) , ..., (x_n, y_n) is $Area = \frac{1}{2} \left[\begin{vmatrix} x_1 & y_1 \\ x_2 & y_2 \end{vmatrix} + \begin{vmatrix} x_2 & y_2 \\ x_3 & y_3 \end{vmatrix} + \begin{vmatrix} x_3 & y_3 \\ x_4 & y_4 \end{vmatrix} + \dots + \begin{vmatrix} x_n & y_n \\ x_1 & y_1 \end{vmatrix} \right]$.
- **Area of triangle bounded by three sides:** If $a_1x + b_1y + c_1 = 0$, $a_2x + b_2y + c_2 = 0$ and $a_3x + b_3y + c_3 = 0$ are the equation of the sides of the triangle, then area of the triangle is given by: $Area = \frac{1}{2|C_1C_2C_3|} \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$,
where $C_1 = a_2b_3 - a_3b_2$, $C_2 = a_3b_1 - a_1b_3$, $C_3 = a_1b_2 - a_2b_1$

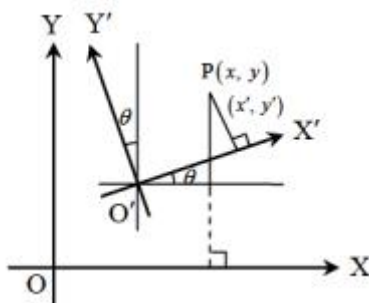
- **Locus:** When a point moves in a plane under certain geometrical conditions, the point traces out a path. This path of a moving point is called its locus.
- **Equation of Locus:** Equation of a locus is the relation which exists between the co-ordinates of any point on the path and which holds for no other point except those lying on the path.
- **Process to find the equation of the locus of a point:**
 - (1) If we are finding the equation of the locus of a point P, assign co-ordinates $P(h, k)$.
 - (2) Express the given conditions as equation in terms of the known quantities to facilitate calculations. We sometimes include some unknown quantities known as parameters.
 - (3) Eliminate the parameters, so that elimination contains only h, k and known quantities.
 - (4) Replace h by x and k by y , in the elimination. The resulting equation would be the equation of the locus of P.
 - (5) If x and y coordinates of the moving point are obtained in terms of third variable t . Eliminate t to obtain the relation in x & y , simplify this relation. This will give the required equation of locus.
- **Translation of axes:** It involves shifting of origin to a new point, the new axes remaining parallel to the original axes. Let OX and OY be the original axes and new origin shifted to $O'(h, k)$. So if old coordinate was (x, y) , then new coordinate is $(x - h, y - k)$.



- **Rotation of Axes without changing origin:** Let OX and OY be the original axes and new axes obtained OX' and OY' by rotating an angle θ in the anticlockwise direction. Let P be any point in the plane with coordinates (x, y) , then new coordinate (x', y') is given by $x' = x \cos \theta + y \sin \theta$, $y' = -x \sin \theta + y \cos \theta$.



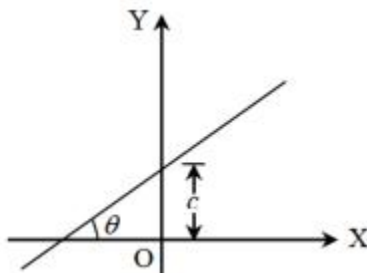
- **Change of origin and rotation of axes:** If origin shifted to $O'(h,k)$ and axes are rotated through an angle θ in the anticlockwise direction. Then new coordinate of (x,y) is (x',y') given by $x' = x \cos \theta + y \sin \theta - h$, $y' = -x \sin \theta + y \cos \theta - k$.



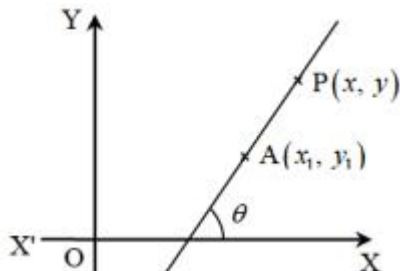
- **General equation of Straight line:** Equation of the form $ax + by + c = 0$, where a, b, c are constant and a, b not simultaneously zero, always represents a straight line.
- **Slope of a line:**
 - (1) If a line makes an angle θ with the positive direction of x axis, then $\tan \theta$ is called as gradient or slope of the line. It is denoted by $m = \tan \theta$.
 - (2) Slope of a line passes through two points (x_1, y_1) and (x_2, y_2) is $\frac{y_2 - y_1}{x_2 - x_1}$.
- **Intercepts of a line on the axes:**
 - (1) **Intercept of a line on x-axis:** If a line cuts x -axis at $(a, 0)$, then a is called as intercept of the line on x -axis and $|a|$ is called as length of intercept of line on x -axis.
 - (2) **Intercept of a line on y-axis:** If a line cuts y -axis at $(0, b)$, then b is called as intercept of the line on y -axis and $|b|$ is called as length of intercept of line on y -axis.
- **Equation of lines parallel to Axes:**
 - (1) Equation of x -axis is $y = 0$.
 - (2) Equation of y -axis is $x = 0$.
 - (3) Equation of a line parallel to x -axis and a unit above x -axis is $y = a$.
 - (4) Equation of a line parallel to x -axis and a unit below x -axis is $y = -a$.
 - (5) Equation of a line parallel to y -axis and a unit right of y -axis is $x = a$.
 - (6) Equation of a line parallel to y -axis and a unit left of y -axis is $x = -a$.

• **Equation of line Various Forms:**

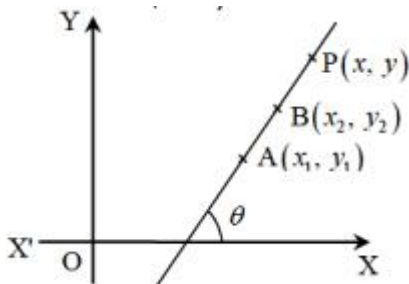
- (1) **Slope Intercept Form:** Equation of straight line whose slope is m and y intercept is c is $y = mx + c$.



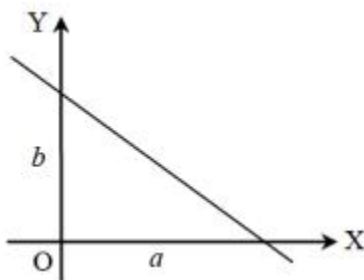
- (2) **Point-Slope Form:** Equation of straight line whose slope is m and passes through a point (x_1, y_1) is $y - y_1 = m(x - x_1)$.



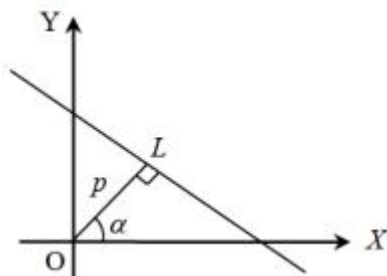
- (3) **Two point Form:** Equation of straight line passes through two points (x_1, y_1) and (x_2, y_2) is $y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$.



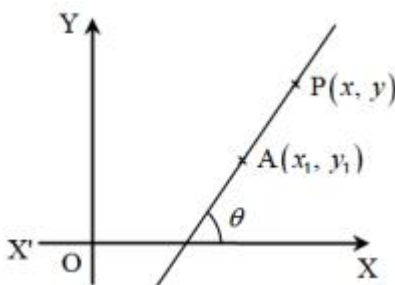
- (4) **Intercept Form:** Equation of straight line which cuts off intercepts a and b on x-axis and y-axis respectively is $\frac{x}{a} + \frac{y}{b} = 1$.



- (5) **Normal Form:** Equation of straight line upon which the length of perpendicular from the origin is p and the perpendicular makes an angle α with the positive direction of x-axis is $x \cos \alpha + y \sin \alpha = p$.



- (6) **Parametric Form or Symmetric Form:** Equation of straight line passing through the point (x_1, y_1) and making an angle q with the positive direction of x-axis is $\frac{x - x_1}{\cos q} = \frac{y - y_1}{\sin q} = r$ where $0 \leq q \leq 2\pi$ and r is the distance of the point (x, y) from the point (x_1, y_1) . The co-ordinate of any point on the line at a distance r from the point (x_1, y_1) can be taken as $(x_1 + r \cos q, y_1 + r \sin q)$ or $(x_1 - r \cos q, y_1 - r \sin q)$ where $0 \leq q \leq \pi$.



- **Reduction of the general equation of straight line to different forms:** Let equation of straight line in general form is $ax + by + c = 0$.

- (1) **Slope Intercept Form:**

$$ax + by + c = 0 \Rightarrow y = \frac{-a}{b}x + \frac{-c}{b}, \text{ where slope} = \frac{-a}{b} \text{ and y intercept} = \frac{-c}{b}$$

- (2) **Intercept Form:**

$$ax + by + c = 0 \Rightarrow ax + by = -c \Rightarrow \frac{a}{-c}x + \frac{b}{-c}y = 1 \Rightarrow \frac{x}{-c/a} + \frac{y}{-c/b} = 1$$

$$\text{Here x intercept} = \frac{-c}{a} \text{ and y intercept} = \frac{-c}{b}.$$

- (3) **Normal Form:**

- (a) **Case 1:**

When $c < 0$ or $-c > 0$, then

$$ax + by + c = 0 \Rightarrow ax + by = -c \Rightarrow \frac{a}{\sqrt{a^2 + b^2}}x + \frac{b}{\sqrt{a^2 + b^2}}y = \frac{-c}{\sqrt{a^2 + b^2}}$$

Which is of the normal form $x \cos \alpha + y \sin \alpha = p$

Where $\cos a = \frac{a}{\sqrt{a^2 + b^2}}$, $\sin a = \frac{b}{\sqrt{a^2 + b^2}}$, $p = \frac{-c}{\sqrt{a^2 + b^2}}$

(b) **Case 2:**

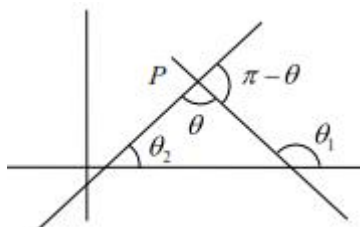
When $c > 0$ or $-c < 0$, then

$$ax + by + c = 0 \Rightarrow ax + by = -c \Rightarrow \frac{a}{-\sqrt{a^2 + b^2}}x + \frac{b}{-\sqrt{a^2 + b^2}}y = \frac{-c}{-\sqrt{a^2 + b^2}}$$

Which is of the normal form $x \cos a + y \sin a = p$

Where $\cos a = \frac{a}{-\sqrt{a^2 + b^2}}$, $\sin a = \frac{b}{-\sqrt{a^2 + b^2}}$, $p = \frac{c}{\sqrt{a^2 + b^2}}$

- **Angle between two intersecting Lines:** The angle between two lines $y = m_1x + c_1$ and $y = m_2x + c_2$ is given by $\tan \theta = \pm \frac{m_1 - m_2}{1 + m_1m_2} \Rightarrow \tan \theta = \left| \frac{m_1 - m_2}{1 + m_1m_2} \right|$, provided no line perpendicular to x-axis.



- (1) If both the lines are perpendicular to x-axis, then angle between them is 0° .
- (2) If any one of the line is perpendicular to x-axis, then the slope of that line is infinite.

Let $m_1 = \infty$, then $\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1m_2} \right| = \left| \frac{1 - \frac{m_2}{m_1}}{\frac{1}{m_1} + m_2} \right| = \left| \frac{1 - 0}{0 + m_2} \right| = \left| \frac{1}{m_2} \right|$.

- (3) Two lines $y = m_1x + c_1$ and $y = m_2x + c_2$ are parallel if $m_1 = m_2$.
 - (4) Two lines $y = m_1x + c_1$ and $y = m_2x + c_2$ are perpendicular if $m_1 \times m_2 = -1$.
- **Condition for two ;one to be coincident, parallel or perpendicular or Intersecting:** Two lines $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ are:

- (1) **Coincident** if: $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$.
- (2) **Parallel** if: $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$.
- (3) **Perpendicular** if: $a_1a_2 + b_1b_2 + c_1c_2 = 0$.
- (4) **Intersecting** if: $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$.

- **Equation of a line parallel to a given Line:** Equation of a line parallel to a given line $ax + by + c = 0$ is $ax + by + K = 0$, where K is any constant.
- **Equation of a line perpendicular to a given Line:** Equation of a line perpendicular to a given line $ax + by + c = 0$ is $ay - bx + K = 0$, where K is any constant.
- **Point of intersection of two given lines:** Point of intersection of two given lines $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ is $\left(\frac{b_1c_2 - b_2c_1}{a_1b_2 - a_2b_1}, \frac{c_1a_2 - c_2a_1}{a_1b_2 - a_2b_1} \right)$.
- **Interior angles of a triangle:** Let slopes three sides of the triangle are in decreasing order as $m_1 > m_2 > m_3$. Then $\tan a = \frac{m_1 - m_2}{1 + m_1m_2}$, $\tan b = \frac{m_2 - m_3}{1 + m_2m_3}$ and $\tan g = \frac{m_3 - m_1}{1 + m_3m_1}$.
- **Equation of Line through the intersection of two lines:** Equation of line passes through the intersection of two lines $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ is $a_1x + b_1y + c_1 + K(a_2x + b_2y + c_2) = 0$.
- **Condition for three lines to be concurrent:** Three lines $a_1x + b_1y + c_1 = 0$, $a_2x + b_2y + c_2 = 0$ and $a_3x + b_3y + c_3 = 0$ are concurrent if $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0$.
- **Position of two point relative to a line:**
 - (1) Two points (x_1, y_1) and (x_2, y_2) are on the same side of the line $ax + by + c = 0$ if $a_1x + b_1y + c_1$ and $a_2x + b_2y + c_2$ are of same sign.
 - (2) Two points (x_1, y_1) and (x_2, y_2) are on the opposite side of the line $ax + by + c = 0$ if $a_1x + b_1y + c_1$ and $a_2x + b_2y + c_2$ are of opposite sign
- **Length of perpendicular distance from a point on a line:** Length of perpendicular distance from a point (x_1, y_1) to the line $ax + by + c = 0$ is $d = \left| \frac{a_1x + b_1y + c_1}{\sqrt{a^2 + b^2}} \right|$.
- **Distance between two parallel lines:** Distance between two parallel lines $ax + by + c_1 = 0$ and $ax + by + c_2 = 0$ is $d = \left| \frac{c_1 - c_2}{\sqrt{a^2 + b^2}} \right|$.

- Area of a parallelogram or a rhombus, equation of whose sides are given:

(1) Area of Parallelogram = $\frac{p_1 p_2}{\sin q}$,

(2) Area of Rhombus = $\frac{p_1^2 p_2}{\sin q}$ as $p_1 = p_2$ in case of rhombus.

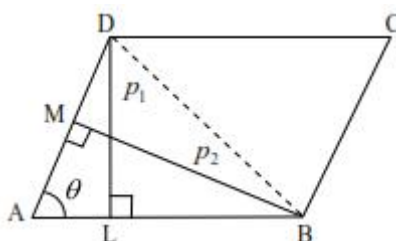
Area of Rhombus = $\frac{1}{2} d_1 d_2$.

$p_1 = DL =$ Distance between the lines AB and CD.

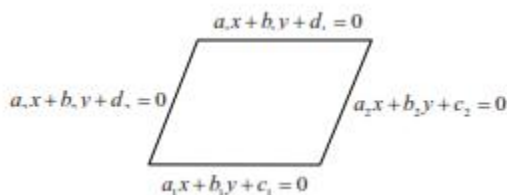
$p_2 = BM =$ Distance between the lines AD and BC.

$q =$ Angle between the adjacent sides AB and AD.

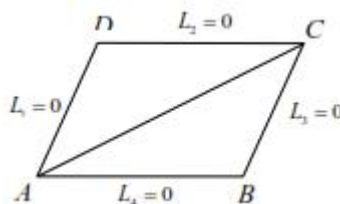
d_1, d_2 are the diagonals



- (3) If the equation of sides of a parallelogram are $a_1x + b_1y + c_1 = 0$, $a_1x + b_1y + d_1 = 0$, $a_2x + b_2y + c_2 = 0$, $a_2x + b_2y + d_2 = 0$, then area is given by $\left| \frac{(c_1 - d_1)(c_2 - d_2)}{a_1b_2 - a_2b_1} \right|$.



- (4) If the equation of sides of a parallelogram are $L_1 = 0$, $L_2 = 0$, $L_3 = 0$, $L_4 = 0$, then the diagonal BD is given by $L_2L_3 - L_1L_4 = 0$ and diagonal AC is given by $L_2L_3 - L_1L_4 = 0$.



- Equation of a straight line passes through a given point and making an angle with another given line: Equation of straight line which passes through a point (x_1, y_1) and

making an angle α with a given line $y = mx + c$ are $y - y_1 = \frac{m + \tan \alpha}{1 - m \tan \alpha}(x - x_1)$ and $y - y_1 = \frac{m - \tan \alpha}{1 + m \tan \alpha}(x - x_1)$.

- **Equation of the bisector of the angle between two lines:** Equation of bisector of angle between two lines $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ are $\frac{a_1x + b_1y + c_1}{\sqrt{a_1^2 + b_1^2}} = \pm \frac{a_2x + b_2y + c_2}{\sqrt{a_2^2 + b_2^2}}$.

- **To find the equation of bisector of the acute and obtuse angle between two lines:** Let the equation of two lines be $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ where $c_1, c_2 > 0$.

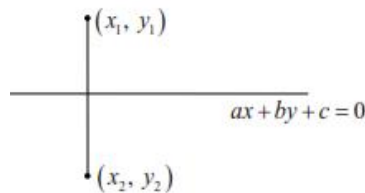
(1) $\frac{a_1x + b_1y + c_1}{\sqrt{a_1^2 + b_1^2}} = + \frac{a_2x + b_2y + c_2}{\sqrt{a_2^2 + b_2^2}}$ is the equation of bisector containing origin.

(2) $\frac{a_1x + b_1y + c_1}{\sqrt{a_1^2 + b_1^2}} = - \frac{a_2x + b_2y + c_2}{\sqrt{a_2^2 + b_2^2}}$ is the equation of bisector not containing origin.

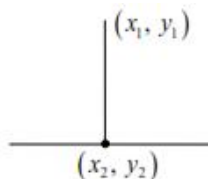
(3) If $a_1a_2 + b_1b_2 > 0$, then the origin lie in obtuse angle and hence $\frac{a_1x + b_1y + c_1}{\sqrt{a_1^2 + b_1^2}} = + \frac{a_2x + b_2y + c_2}{\sqrt{a_2^2 + b_2^2}}$ is the obtuse angle bisector.

(4) If $a_1a_2 + b_1b_2 < 0$, then the origin lie in acute angle and hence $\frac{a_1x + b_1y + c_1}{\sqrt{a_1^2 + b_1^2}} = + \frac{a_2x + b_2y + c_2}{\sqrt{a_2^2 + b_2^2}}$ is the acute angle bisector.

- **Image of a point:** The co-ordinate of the image (x_2, y_2) of a given point (x_1, y_1) in the line $ax + by + c = 0$ is given by $\frac{x_2 - x_1}{a} = \frac{y_2 - y_1}{b} = \frac{-2(ax_1 + by_1 + c)}{a^2 + b^2}$.

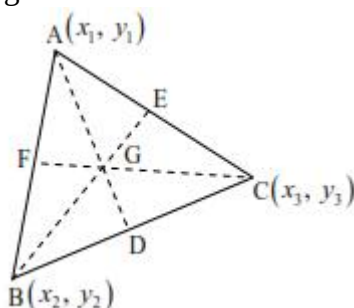


- **Foot of the perpendicular:** The co-ordinate of the foot of the perpendicular (x_2, y_2) of a given point (x_1, y_1) on the line $ax + by + c = 0$ is given by $\frac{x_2 - x_1}{a} = \frac{y_2 - y_1}{b} = \frac{-(ax_1 + by_1 + c)}{a^2 + b^2}$.

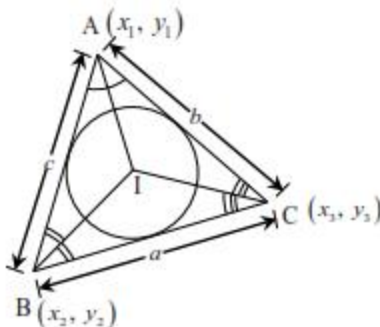


• **Standard Points of a triangle:**

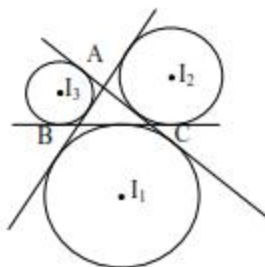
- (1) **Centroid of a triangle:** The point of intersection of medians of the triangle is called as centroid of the triangle.



- (a) Centroid divided the median in the ratio 2:1.
 (b) Co-ordinate of centroid of a triangle with vertices (x_1, y_1) , (x_2, y_2) and (x_3, y_3) is $\left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$.
 (c) If P is any point in the plane of the triangle ABC and G is the centroid then $PA^2 + PB^2 + PC^2 = GA^2 + GB^2 + GC^2 + 3PG^2$.
 (d) If G is the centroid of the triangle ABC, then $AB^2 + BC^2 + CA^2 = 3(GA^2 + GB^2 + GC^2)$
- (2) **In-centre of a triangle:** The point of intersection of angle bisector of the triangle is called as in-centre of the triangle.



- (a) Co-ordinate of in-centre of a triangle with vertices (x_1, y_1) , (x_2, y_2) and (x_3, y_3) is $\left(\frac{ax_1 + bx_2 + cx_3}{a + b + c}, \frac{ay_1 + by_2 + cy_3}{a + b + c} \right)$.
- (3) **Ex-centre of a triangle:** A circle touches one side outside the triangle and other two extended sides, then the circle is called as Ex-circle and its centre is called as ex-centre. Let ABC is a triangle with three ex-circles with their centres I_1, I_2, I_3 opposite to the vertices A, B and C. If vertices of the triangle ABC are $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$, then:

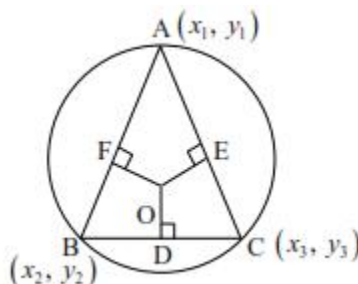


$$(a) \quad I_1 = \left(\frac{-ax_1 + bx_2 + cx_3}{-a + b + c}, \frac{-ay_1 + by_2 + cy_3}{-a + b + c} \right).$$

$$(b) \quad I_1 = \left(\frac{ax_1 - bx_2 + cx_3}{a - b + c}, \frac{ay_1 - by_2 + cy_3}{a - b + c} \right).$$

$$(c) \quad I_1 = \left(\frac{ax_1 + bx_2 - cx_3}{a + b - c}, \frac{ay_1 + by_2 - cy_3}{a + b - c} \right).$$

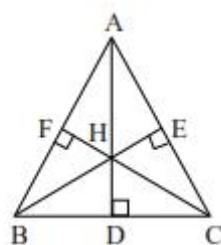
- (4) **Circum-centre of a triangle:** The point of intersection of perpendicular bisector of sides of a triangle is called as circum-centre.



- (a) If vertices of the triangle ABC are $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$, then its circum-centre is given by:

$$\left(\frac{x_1 \sin 2A + x_2 \sin 2B + x_3 \sin 2C}{\sin 2A + \sin 2B + \sin 2C}, \frac{y_1 \sin 2A + y_2 \sin 2B + y_3 \sin 2C}{\sin 2A + \sin 2B + \sin 2C} \right)$$

- (5) **Ortho-centre of a triangle:** The point of intersection of altitudes of a triangle from the vertices to its opposite sides is called as ortho-centre.



- (a) If vertices of the triangle ABC are $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$, then its ortho-centre is given by:

$$\left(\frac{x_1 \tan A + x_2 \tan B + x_3 \tan C}{\tan A + \tan B + \tan C}, \frac{y_1 \tan A + y_2 \tan B + y_3 \tan C}{\tan A + \tan B + \tan C} \right).$$

- (b) If H is the orthocenter of triangle ABC, then C is the orthocenter of triangle HAB, B is orthocenter of triangle HAC and A is the orthocenter of triangle HBC.

- (c) If any two sides out of three sides of a triangle are perpendicular, then orthocenter is the point of intersection of two perpendicular sides.
- (d) In right angled triangle vertex opposite to hypotenuse is the orthocenter.
- (e) In a right triangle midpoint of the hypotenuse is the circum-centre.

